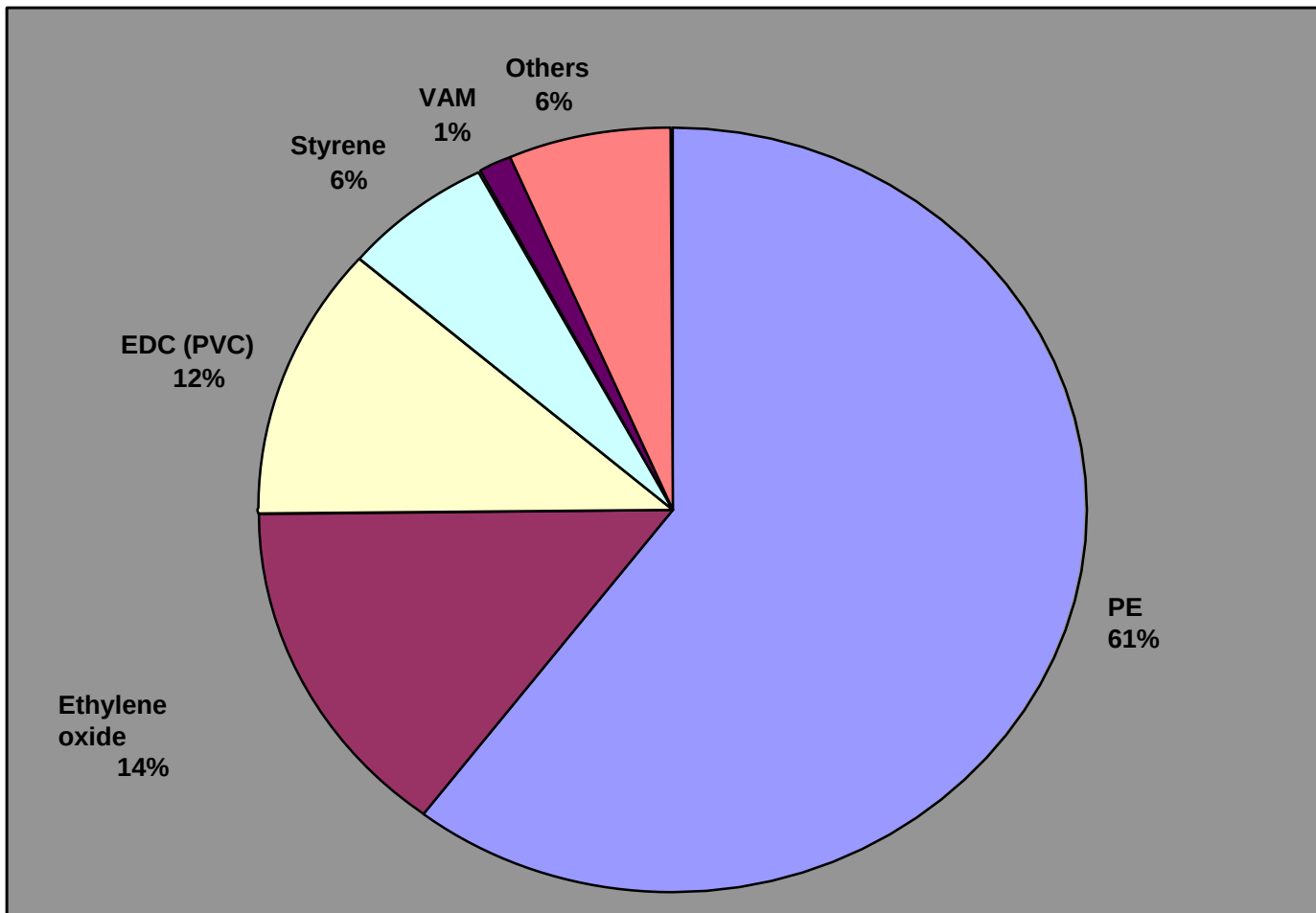


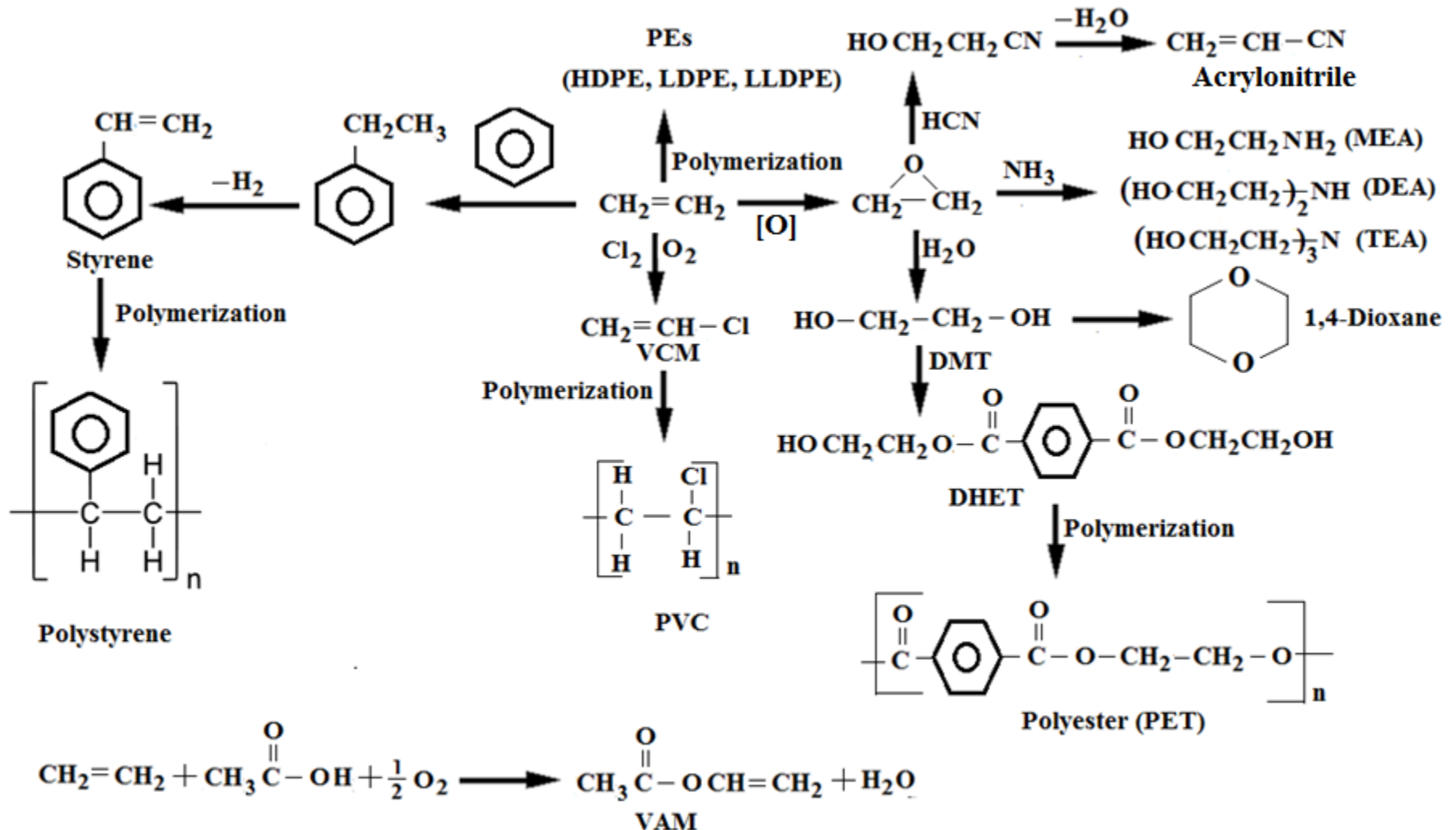
Petrochemical Technology

Chemicals from Ethylene

Ethylene Consumption Patterns



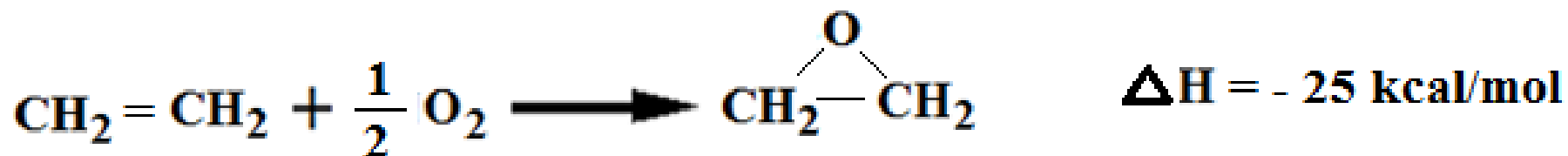
Major Chemicals from Ethylene



Vinyl Acetate Monomer (VAM) is the precursor to **polyvinyl acetate (PVA)**. It is an intermediate used in the manufacture of a number of industrial polymers and resins for the production of adhesives, coatings, paints, films and textiles, as well as wire and cable insulation.

Ethylene Oxide

Ethylene Oxide (EO) is the second largest consumer of Ethylene. EO finds key applications in manufacturing Ethylene Glycol, the most common of which is monoethylene glycol (MEG). MEG is used for making PET resins which is used for making bottles and is also used in several antifreeze applications. Demand for several EO derivatives like ethoxylates (used in shampoo, kitchen cleaners, etc.), glycol ethers (solvents, fuels, etc.) and ethanolamines (surfactants, personal care products, etc.) will always remain healthy.



Catalyst: $\text{Ag}_2\text{O}/\text{Al}_2\text{O}_3$

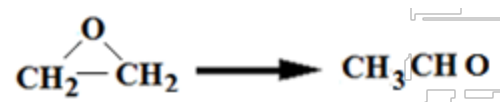
[Catalyst contains 10-15% of silver and promoters-compounds of alkalis or alkali earth metals, especially Cs and Ba.]

Oxidation with air : **260-280°C**; Oxidation with oxygen: **230°C**

Pressure: **10-30 atm**

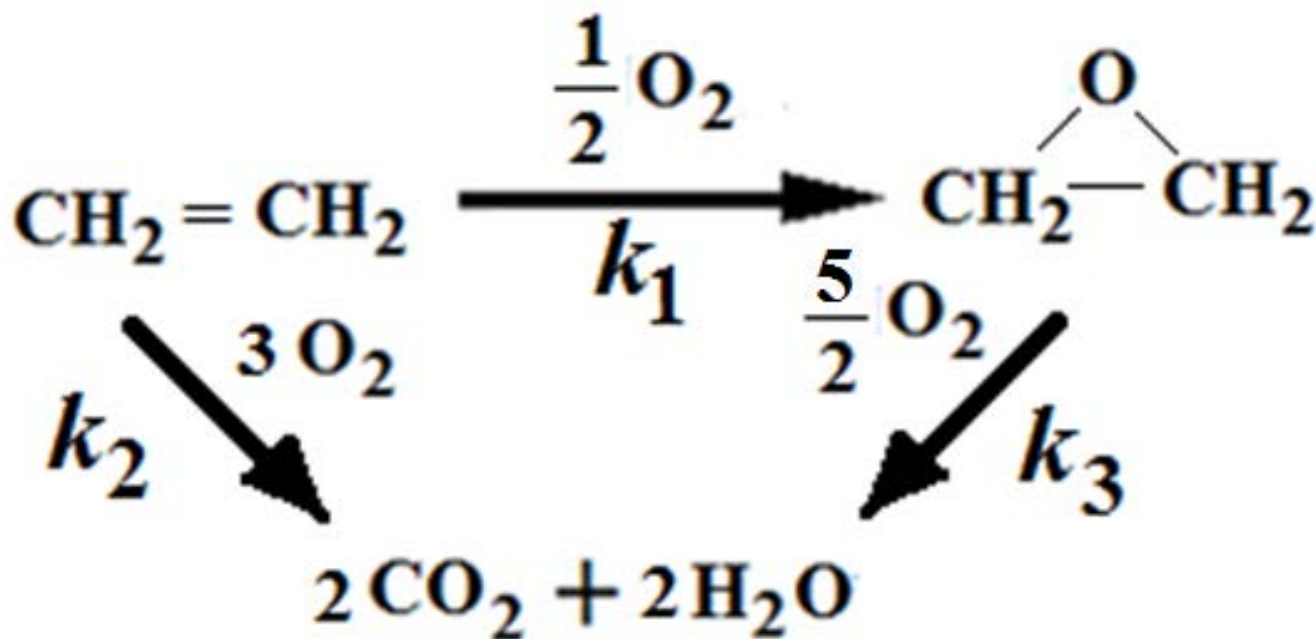
High-purity, sulphur-free ethylene is required to avoid catalyst poisoning.

Ethylene concentration is about 20-30 vol% or 5 vol% when oxygen or air is used as the oxidant, respectively. The main byproducts are CO_2 and H_2O , and a very small amount of **acetaldehyde** is formed via isomerization of ethylene oxide.

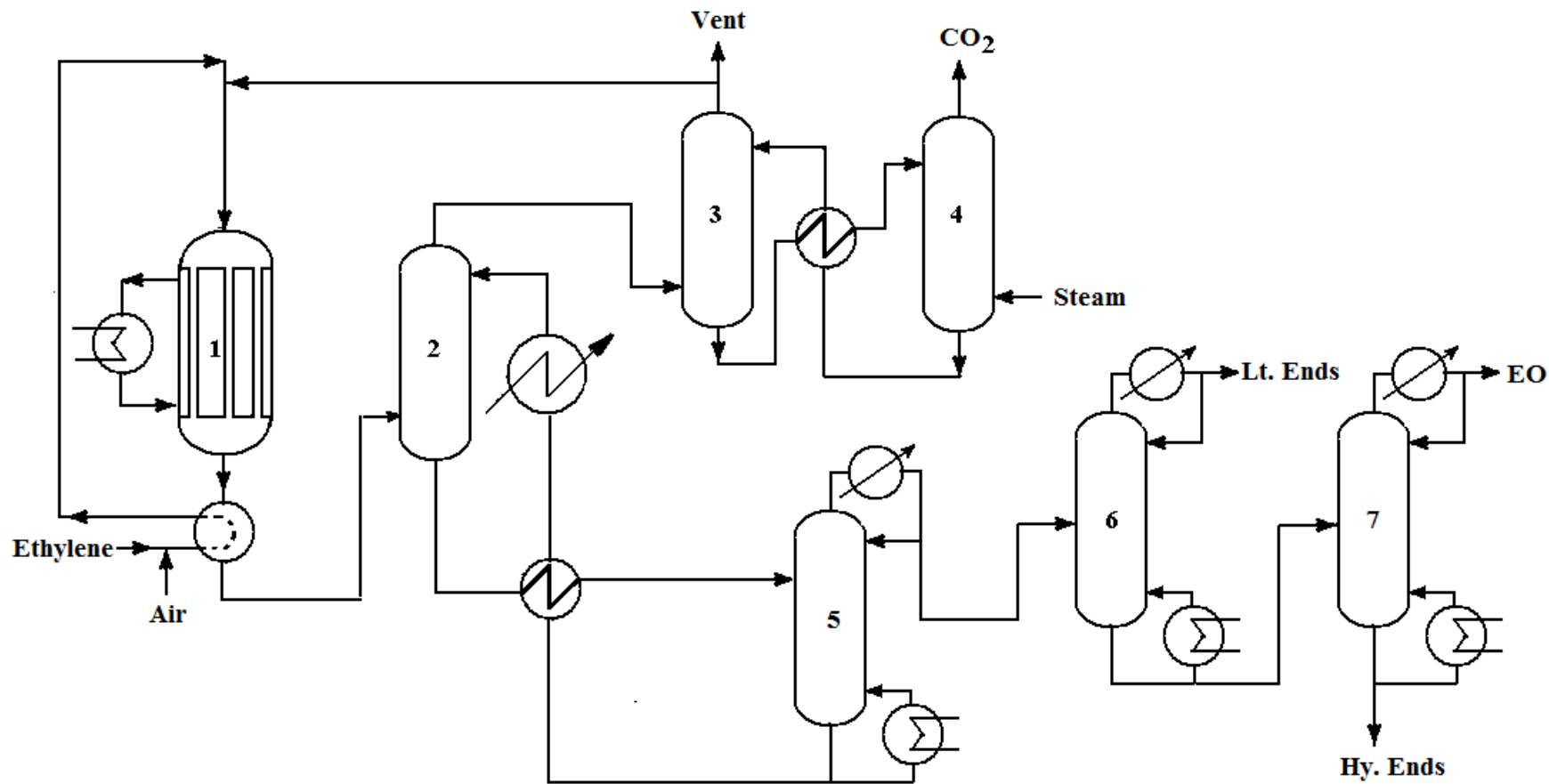


Selectivity to ethylene oxide is 65-75% (air process) or 70-80% (O₂ process).

Selectivity is improved by the addition of chlorine compounds (such as ethylene dichloride, ethyl and vinyl chloride), which reduce the production of byproduct CO₂ and ensure an even silver surface coating.



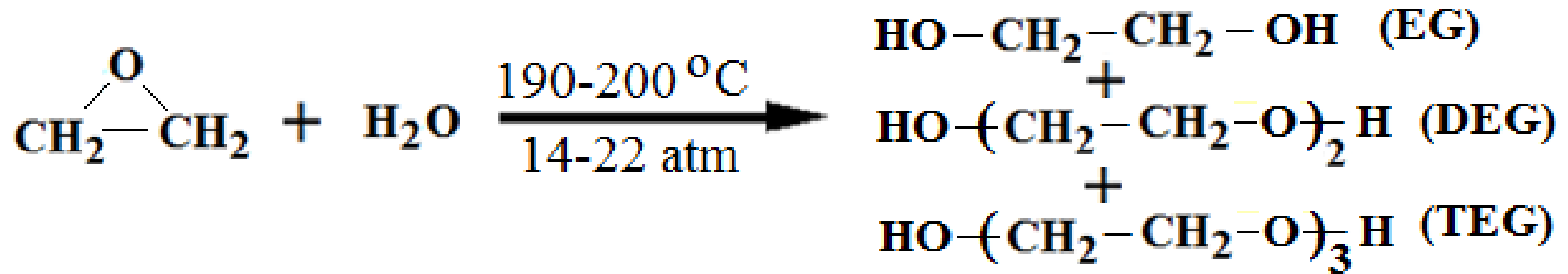
On the best industrial catalysts k_1/k_2 is 6.0
and k_2/k_3 is 2.5.



1: Reactor; 2: EO Absorber; 3: CO₂ Absorber; 4: CO₂ Stripper; 5: EO Stripper; 6: Lt. Ends Removal Column; 7: EO Column

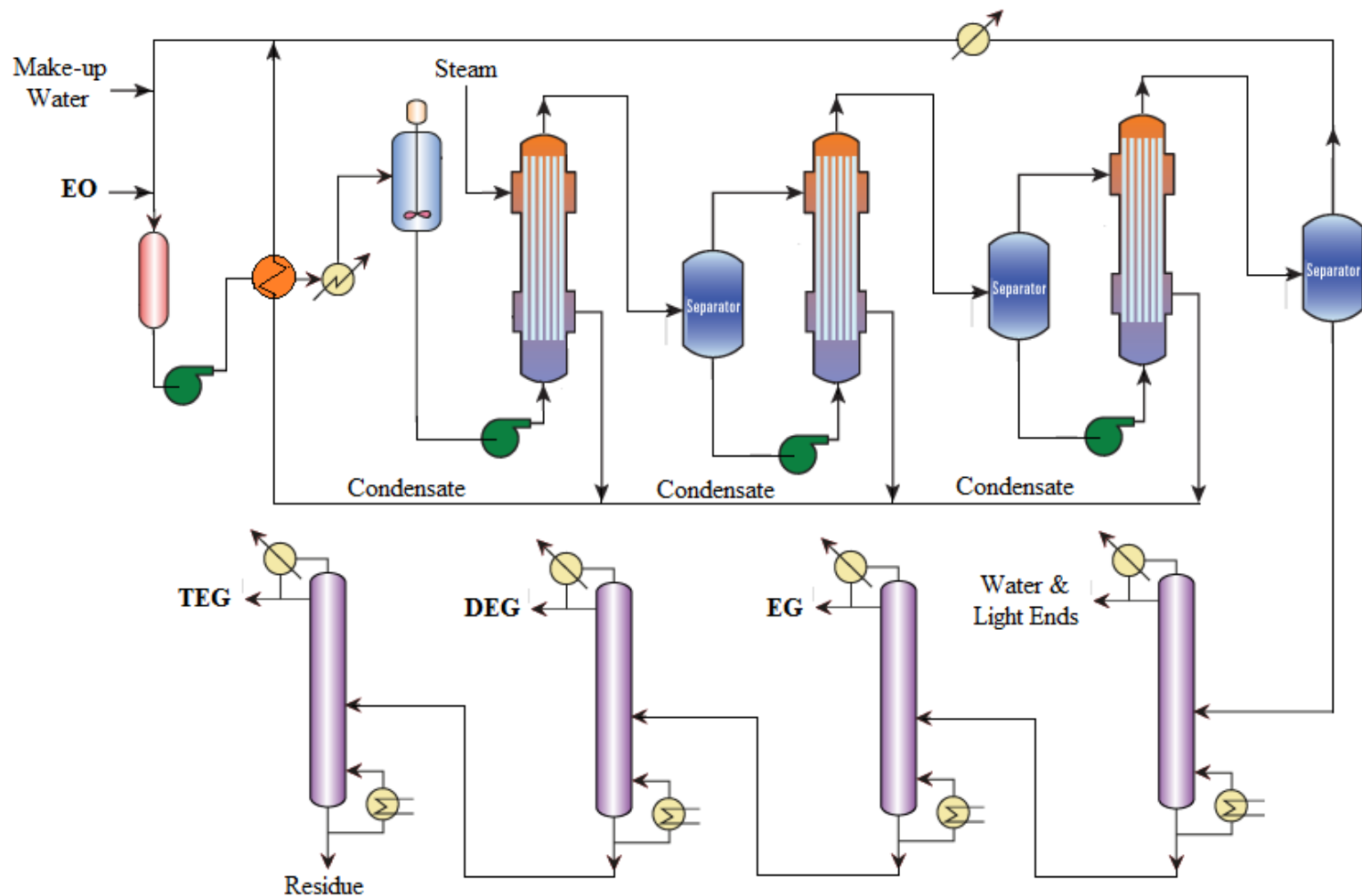
Flowsheet for the production of Ethylene Oxide (EO)

Ethylene Glycol



A huge excess of water is used (18-24 moles of water/mole of EO) to prevent formation of di-, tri- and higher ethylene glycols.

Even with high water/EO ratio, mono- , di- and tri-ethylene glycols are obtained in a weight ratio of 91 : 8.6 : 0.4.

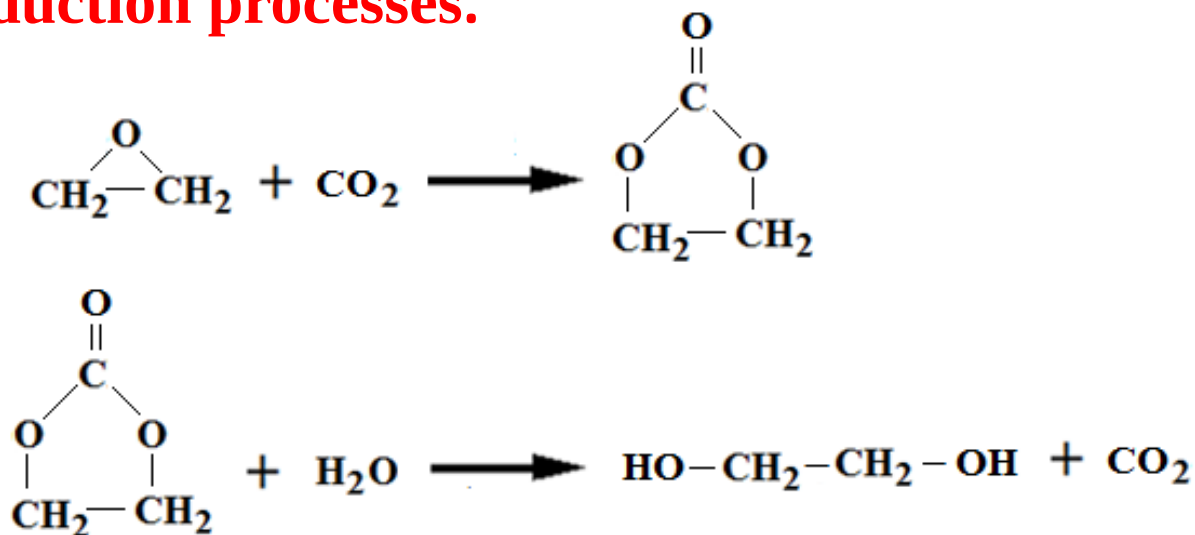


Flowsheet for the production of Ethylene Glycol

Monoethylene Glycol from EO

Ethylene oxide is reacted with CO_2 , forming ethylene carbonate, which is then hydrolyzed to form MEG and CO_2 . Both reactions are carried out in the liquid phase using homogeneous catalysts.

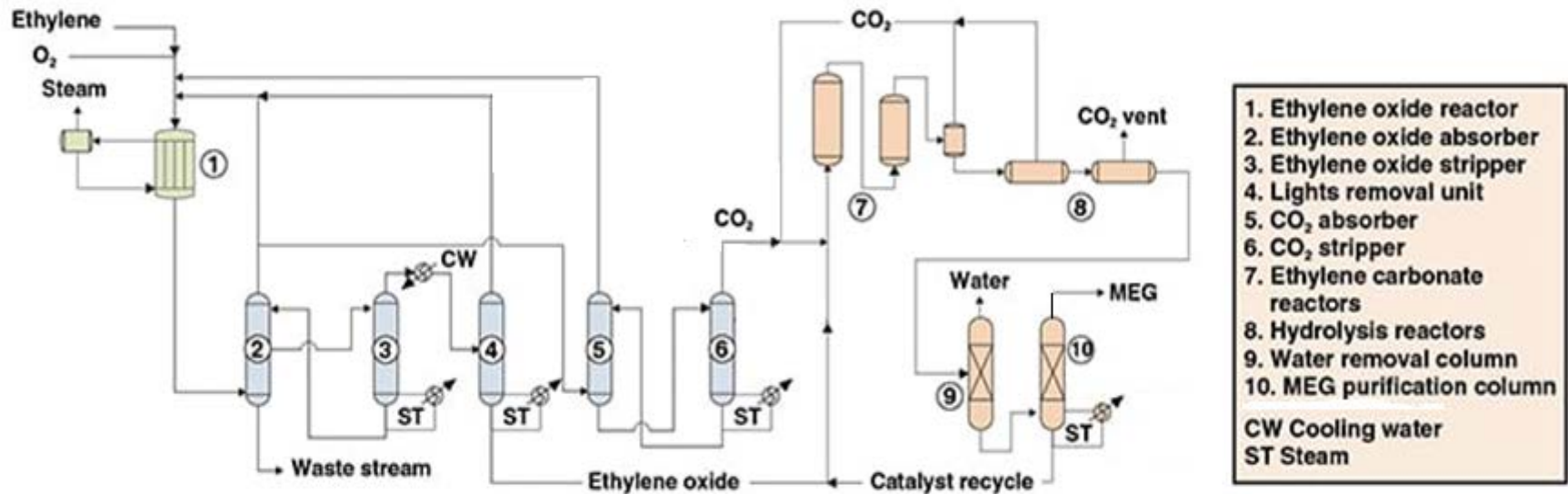
An important feature of the process is the negligible production of diethylene glycol (DEG) and triethylene glycol (TEG), which occur as byproducts in other ethylene glycol production processes.



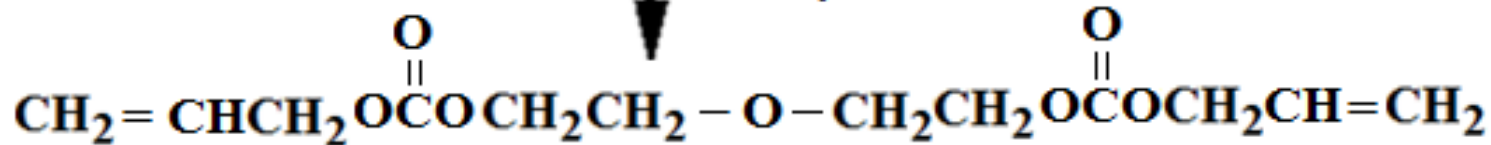
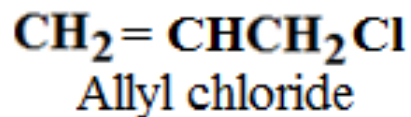
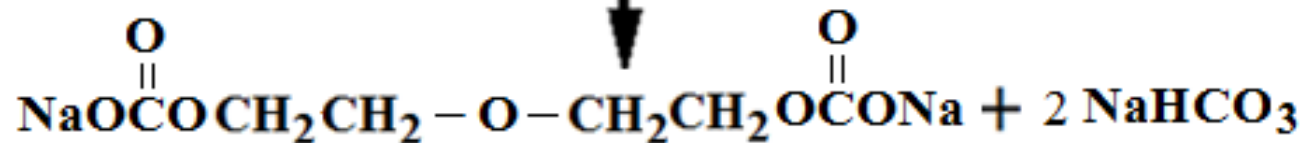
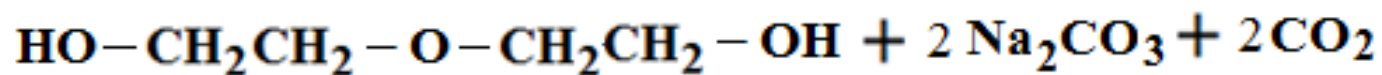
Shell's OMEGA Process

(OMEGA stands for “Only MEG Advantage”)

Shell's OMEGA process for producing mono-ethylene glycol (MEG) from ethylene in a 2-stage, fully catalytic process, which has enabled many operators to achieve a conversion efficiency of over 99% compared to around 90% for conventional processes which utilise a thermal route to convert ethylene oxide (EO) into MEG.



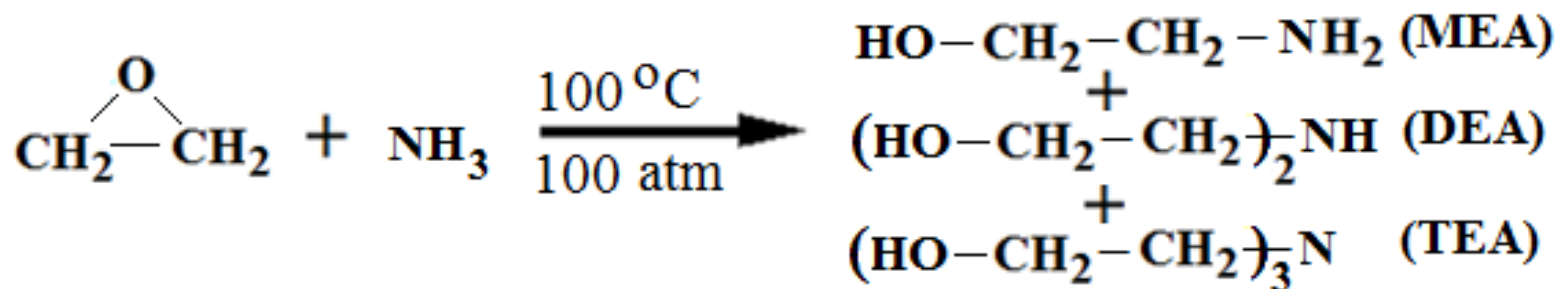
Flowsheet for the production of Monoethylene Glycol (OMEGA)



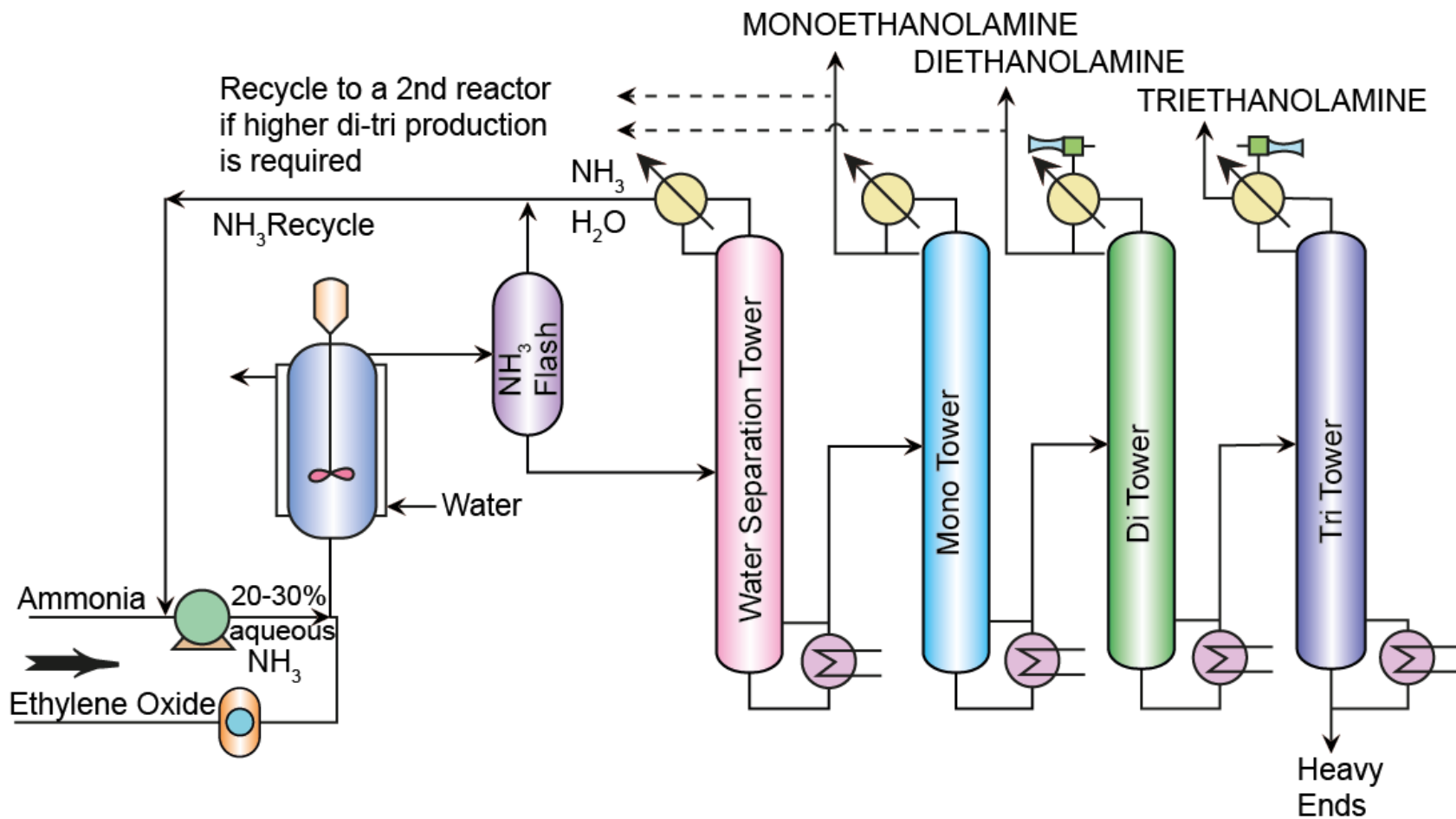
Diethylene glycol bis(allyl carbonate)
Allyl diglycol carbonate

Diethylene glycol bis(allyl carbonate) is a specialty polycarbonate polymer whose major use is for molding eyeglass lenses by *in situ* polymerization.

Ethanol Amines

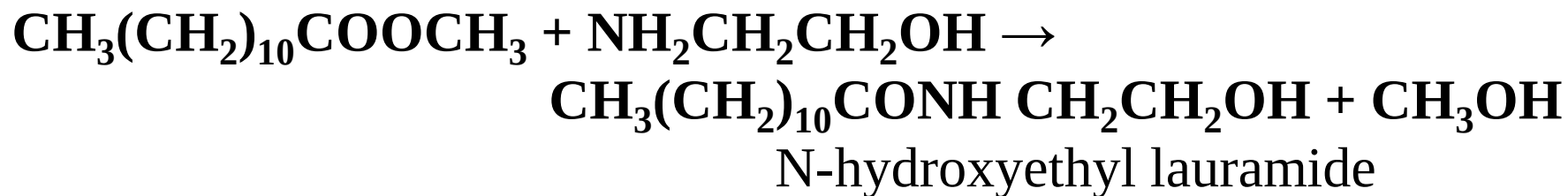


$\text{NH}_3/\text{EO} = 0.5 - 3.0,$
Temperature = $35 - 275^\circ\text{C},$
Pressure = $1 - 100 \text{ atm}$

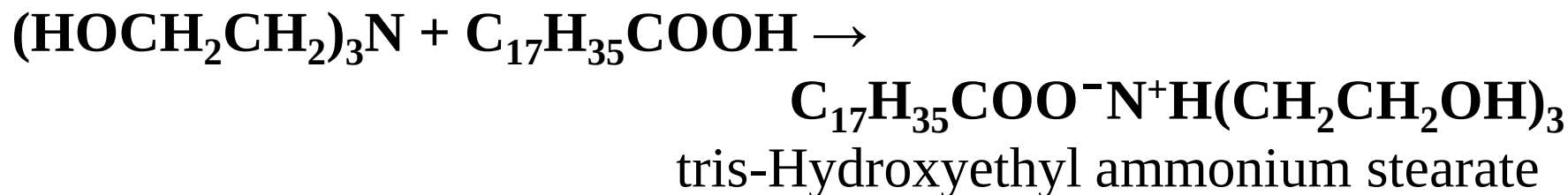


Flowsheet for the production of ethanol amines from ethylene oxide

The reaction of fatty acid methyl esters such as methyl laurate with ethanol amines provides fatty ethanol amides, widely used as foam stabilizers in detergent formulations.



Of the three ethanol amines, triethanol amine enjoys about 40% of total demand, its major use being the formation of a quaternary salt with fatty acids to provide a surfactant useful in dry cleaning and cosmetic formulations.



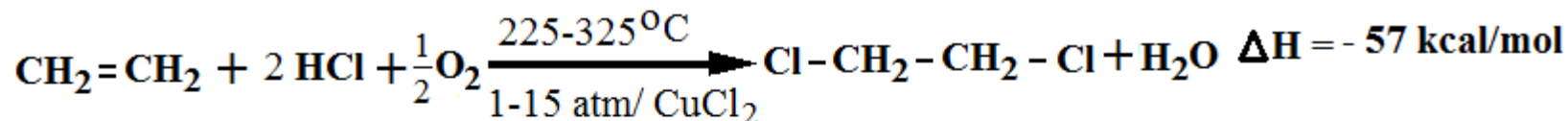
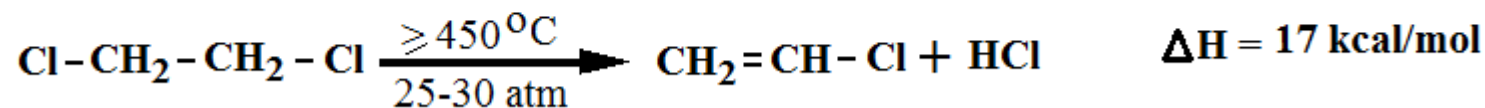
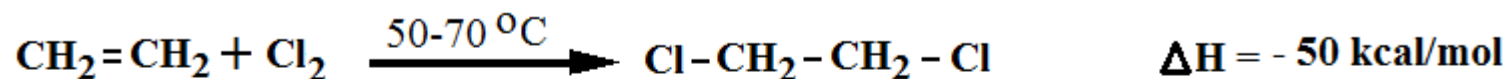
Vinyl Chloride Monomer (VCM)

Global Vinyl Chloride Monomer (VCM) Market was valued at **USD 83.6 billion in 2023** and is **expected to reach USD 164.2 billion by 2032**, registering a CAGR of 7.7% during the forecast period of 2024-2032.

The global VCM industry capacity was **52.38 mtpa in 2022**.

Over 99% of total global VCM consumption is used for PVC production.

Balanced Chlorination-Oxychlorination process:



Mechanism of Oxychlorination

Copper chloride is universally applied as catalyst. Known as the modified Deacon catalyst, CuCl_2 is supported on alumina and contains KCl.

Under operating conditions, a CuCl_2 - Cu_2Cl_2 -KCl ternary mixture, possibly in the molten state, is the active catalyst.

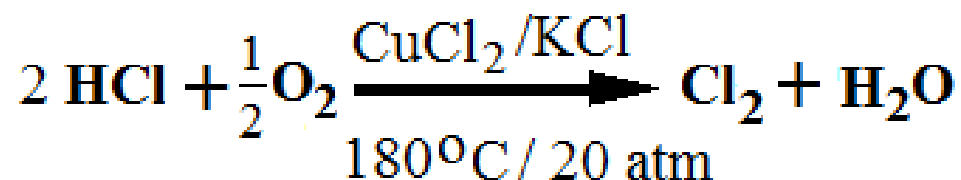
CuCl_2 is believed to be the active chlorinating agent producing EDC directly without the involvement of elemental chlorine.

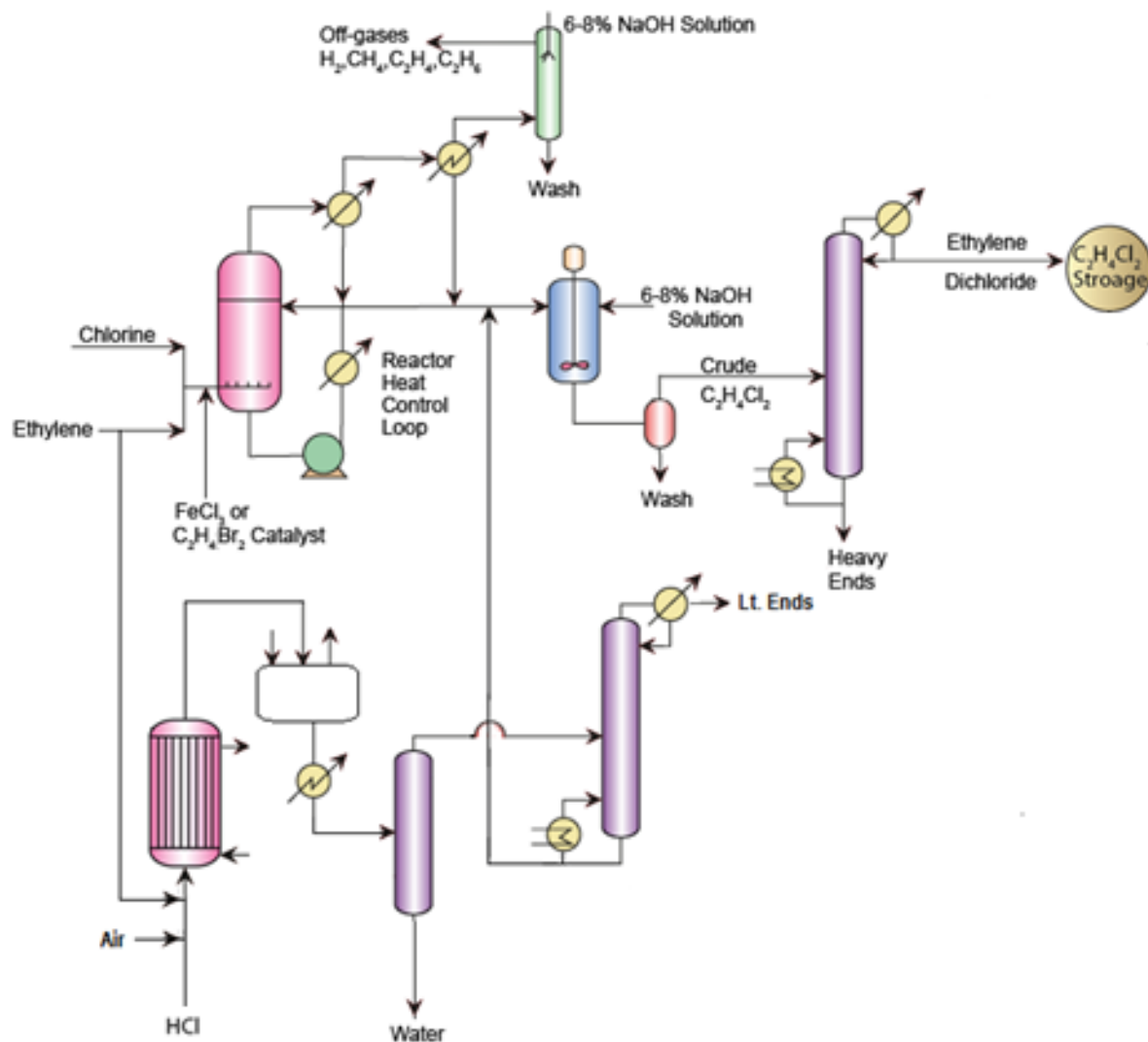
The coupled oxidoreduction steps are indicated below:



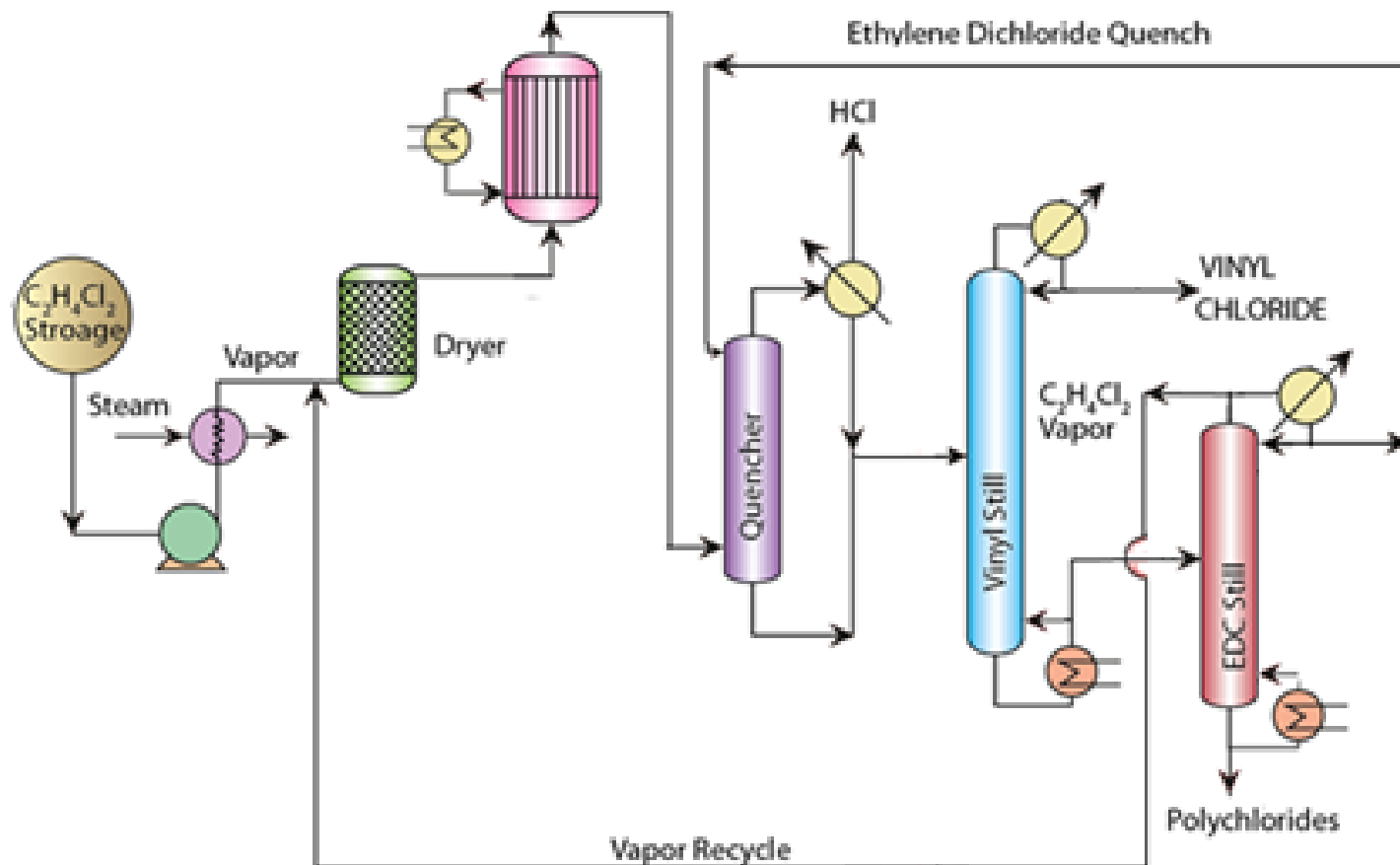
Kellog's technology for oxidation of HCl to Cl₂

A modified vinyl chloride process combines Monsanto's direct chlorination, cracking, and purification technologies and Kellog's proprietary technology for oxidation of HCl to Cl₂. In this process, the inefficient hazardous oxychlorination step is eliminated. Instead chlorine is recovered after the oxidation of HCl and used again in the direct chlorination of ethylene. This improved technology is more economical and less polluting and yields high purity vinyl chloride.





Flowsheet for the Production of Ethylene Dichloride (EDC) from Ethylene



Flowsheet for the Production of Vinyl Chloride Monomer (VCM) from Ethylene Dichloride (EDC)

Thank you!